

STAT8017 DATA MINING TECHNIQUES

Department of Statistics & Actuarial Science
University of Hong Kong
(First Semester, 2026-2027)

Teaching Team:

- Dr. LI, Guodong (Instructor)
Office: RRS222; Email: gdli@hku.hk; Tel: 3917 1986
- Tutors:
Mr. YU, Shibo – Email: shibo-yu01@connect.hku.hk;
Mr. YU, Liting – Email: lityu@connect.hku.hk.

Teaching Plan:

- Lecture & Lab: 7:00-9:50pm on Wednesday at T2, Meng Wah Complex; a separate tutorial from 8:00-9:50pm on Wednesday at the same classroom;
- There is no class on Oct. 14, and the last day is on Nov. 25.

Course Description:

With the rapid developments in computer and data storage technologies, the fundamental paradigms of classical data analysis are mature for change. Data mining techniques aim at helping people to work smarter by revealing underlying structure and relationships in large amounts of data. This course takes a practical approach to introduce the new generation of data mining techniques and show how to use them to make better decisions. Topics include data preparation, feature selection, association rules, decision trees, bagging, random forests and gradient boosting, cluster analysis, neural networks, introduction to text mining.

Expected Learning Outcomes:

Students who successfully complete this course will be able to:

CLO1 - Implement data mining process summarized in the acronym SEMMA: sampling, exploring, modifying, modeling, and assessing data.

CLO2 - Understand and apply a wide range of data mining techniques, and recognize their characteristics, strengths and weaknesses.

CLO3 - Be proficient with computer software for data mining (Python library).

CLO4 - Identify and use appropriate data mining techniques for a data mining project, taking into account both the nature of the data to be mined and the goals of the user of the discovered knowledge.

CLO5 - Evaluate the quality of discovered knowledge, taking into account the requirements of the data mining task being solved and the goals of the user.

Teaching and Learning Activities:

1. Lectures:

Lecture notes with overviews of topics will be uploaded to Moodle. This will assess CLOs 1, 2 and 4.

2. Computer Labs:
Students practice concepts learnt in class through some case studies using the Python library in your own laptops. You can also use the computer labs located at Run Run Shaw Building: Room 101 and Room 103. This will assess CLOs 1, 2, 3 and 4.
3. Group Project Work:
See the assessment below for more details. This will assess CLOs 1-5.
4. In-Class Learning Activities:
Students are required to attend the lectures. Active participation in class is encouraged. This will assess CLOs 1, 2 and 4.

Assessment:

1. Assignments (30%)
There will be 3 homework assignments (done individually). Some questions may require you using Python to analyze a data set. Your computer output should be clearly labelled, and edited to remove non-essential portions. Do not simply tack on a pile of output through which the reader will have to read slowly to find the relevant analyses.

Partially or wholly copied assignments will be penalized and/or reported as plagiarism (See university website: <http://www.hku.hk/plagiarism>). This will assess CLOs 2 and 3.

2. Class Tests (40%)
There will be 2 tests conducted in-class (dates to be notified). This will assess CLOs 2-3. If for any reason you are or have been unable to attend a class test, and if you wish to have a supplementary test,
 - a. all full-time students (including MSTAT/MDASC students) should write to the General Office of the School of Computing and Data Science giving reasons for your absence;
 - b. all part-time students should write to the course instructor giving reasons for your absence, within 7 days of the absence.

A special/supplementary test is normally granted to those absent from the original test due to illness and with original medical certificate provided. Students absent due to other reasons are not granted a special/supplementary test unless with very special circumstances and with valid documentary proofs provided.

3. Group Project (30%)
Students should work on the project in teams of **4 to 6** students. Students consider a real problem involving class topics on a data set of interest. Students are encouraged to find their own source of data for the project from their workplace, government records or elsewhere. Each team needs to submit a project proposal, gather the necessary data, carry out the programming and data analysis, and submit a report. Further details will be announced later. This will directly assess CLOs 1-5.

Course Content & Tentative Class Schedule:

1. Introduction to Data Mining (week 1)
2. Feature Engineering and Selection (week 2)
3. Supervised Learning
 - Decision Tree (week 3)
 - Predictive Model Performance Evaluation (week 4)
 - Ensemble Methods (week 5)
 - Support Vector Machines (week 6)
 - Naive Bayes (week 8)
 - Neural Networks and Intro to Deep Learning (week 9)
4. Unsupervised Learning
 - Cluster Analysis (week 7)
 - Association Rules (week 11)
5. Text Mining (week 10)

Reference Books:

- Tan, P.N., Steinbach, M. Karpatne, A. and Kumar, V. (2019). Introduction to Data Mining. 2nd edition, Pearson. (006.312 T1 1st ed.) [1st ed., HKU Lib e-book]
- Hastie, T, Tibshirani, R., and Friedman, J., (2009). The Elements of Statistical Learning: Data Mining, Inference, and Prediction, 2nd edition, Springer, New York. [HKU Lib e-book]
- Bishop, C.M. (2006). Pattern Recognition and Machine Learning, Springer. (006.4 B62 p)
- Murphy, K. P. (2012). Machine Learning: A Probabilistic Perspective, The MIT Press. (006.31 M978 m14)
- Han, J., Kamber, M. and Pei, J. (2012). Data Mining: Concepts and Techniques, 3rd edition, Morgan Kaufmann. [HKU Lib e-book]
- Zhai, C. and Massung, S. (2016). Text Data Management and Analysis: A Practical Introduction to Information Retrieval and Text Mining. ACM Books. [HKU Lib e-book]
- Layton, R. (2017). Learning Data Mining with Python. Second Edition, Packt. [github]
- Kane, F. (2017). Hands-On Data Science and Python Machine Learning. Packt. [HKU Lib e-book] [github]
- Geron, A. (2017). Hands-On Machine Learning with Scikit-Learn & TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems. O'Reilly Media. [HKU Lib e-book] [github]
- Sarkar, D., Bali, R. and Sharma, T. (2018). Practical Machine Learning with Python. Apress. [HKU Lib e-book] [github]
- Raschka, S. and Mirjalili, V. (2017). Python Machine Learning: Machine Learning and Deep Learning with Python, scikit-learn and TensorFlow. Second Edition, Packt. [HKU Lib e-book] [github]